

Research paper

Optimizing research on flat-plate solar collector based on field synergy theory

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ABSTRACT

Flat-plate solar collectors are the leading product in the production of low-temperature hot water market, and are also one of the most widely used technologies for utilizing solar energy to generate low-temperature heat. In this study, a specific type of flat-plate collector optimized using field synergy theory was introduced, with the main change being the use of S-bend structure instead of traditional direct current channel. The study applied numerical simulation to analyze and evaluate the performance of the collector. The results showed that introducing an S-bend with a curvature radius of 90 mm can improve the efficiency of a single block collector by 0.1 %, reduce the temperature of the absorption plate by 0.33°C, and drive the angle between the velocity vector and temperature gradient (β angle) close to zero. Compared with the relatively uniform vortex structure observed in the cavity of the straight tube collector, the S-bend structure generates more complex and irregular vortex patterns. It is worth noting that the introduction of S-bend produces a unique butterfly shaped temperature distribution on the glass surface. The changes that occur in the cavity area are driven by the temperature difference between the upper and lower plate surfaces, and these new phenomena induced by S-bending deserve further in-depth discussion. These findings emphasize the importance of continuous research to develop optimized designs, improve efficiency, and promote wider applications, and provide certain ideas and references for subsequent work.

1. Introduction

The mitigation of the global climate crisis requires rapid reduction of fossil energy use to reduce the environmental damage (Ziming et al., 2020), global warming (Zheng et al., 2016), extreme weather, etc. Solar energy as a renewable energy source is widely perceived to have an important future role in the energy transition needed (Zhang et al., 2018; Luo et al., 2022). Solar energy can be harnessed e.g., through photovoltaics technologies or photo-thermal conversion employing solar collectors to produce heat (Ceron et al., 2015; Jiandong et al., 2015). Solar heating for households and buildings is used in many countries and the system layout is quite simple comprising of e.g. flat plate collectors and a short-term heat storage including an auxiliary heat unit. Improving the collector efficiency reduces in addition to the cost of the installation also the space needed for the collectors.

In recent years, extensive research has been dedicated to flat plate

solar collectors, with a primary focus on enhancing optical efficiency, improving heat transfer within each component, and minimizing overall heat loss. Wherein, enhancing heat transfer within individual components has emerged as a prominent area of investigation, warranting special attention in this article. By delving into these strategies, researchers aim to unlock new frontiers in solar energy utilization, ultimately driving advancements in sustainable energy technologies. Through targeted efforts to enhance heat transfer mechanisms, we move closer to realizing the full potential of flat plate solar collectors in meeting global energy demands while reducing environmental impact. Enhanced heat transfer can be pursued through various approaches, including the optimization of absorbent coatings and endothermic elements, utilization of nanofluids, modification of the shape of absorbing heat plates and internal materials, and application of low-frequency vibrations (Duffie and Beckman, 1980). The study by Reyna Dianela Bacelis-Martínez et al. (2023) presented the synthesis and

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characterization of nickel–cobalt mixed metal oxides employed as the active phase in selective paints for solar absorber coatings on residential flat collectors. The synthesized nickel–cobalt mixed oxides crystallized in the cubic phase associated with the spinel structure, demonstrating excellent thermal stability and reproducibility. Yuriy Selikhov et al. (2022) employed polymeric materials in producing absorbing elements for solar collectors, thereby reducing material consumption and lowering the cost of solar installations. Solar collectors equipped with absorbers made from special-grade polyethylene film are not only highly efficient but also four times more cost-effective compared to collectors with aluminum metal absorbers. Ali Omran Al-Sulttani et al. (2022) developed Graphene nanofluids (T-Gr) to enhance collector efficiency. Various mass concentrations (0.1 %-wt., 0.075 %-wt., 0.050 %-wt., and 0.025 %-wt.) were mixed with ethylene glycol (EG) and distilled water (DW) in different ratios to assess thermal efficiency. Manuel Oliveira et al. (2023) embarked on modeling, simulating, and investigating the impact of modifying the geometry of absorber plates in conventional solar collectors designed for heating, by incorporating V-shaped corrugations. This process involves subsequent adjustments to geometric and optical parameters, ultimately enhancing equipment efficiency. J. Castro et al. (2022); Zheng et al., (2024) optimized the flat plate solar collector for energy efficiency by integrating transparent insulation materials (TIM) composed of plastic and aerogel-silicone. Furthermore, the addition of a layer of aerogel-silica provides enhanced protection against overheating, as it helps regulate the temperature of the plastic TIM. Gao et al. proposed a flat plate solar collector with transparent aerogel layer and back shield plate. The vacuum aerogel flat plate solar collector with optimized structure has ultra-high thermal efficiency in the middle and low temperature range, which is 20.68 % higher than the prototype (Gao et al., 2022).

Although research on the optimization of flat plate collectors has gradually deepened in recent years, and researchers have also made many cavity related optimizations, there is relatively little attention paid to the impact of bent pipes on heat transfer (traditional designs mainly use straight pipes (Duffie and Beckman, 1980)). In this study, we introduce a novel approach employing S-bend tubes to create a curved tube flat plate collector based on the field synergy theory. Through numerical modeling, we investigate the collector's efficiency, flow dynamics within the tube, factors contributing to enhanced heat transfer, and the distribution and movement of air within the cavity between the plate and top glass. This examination aims to evaluate the potential efficiency gains achievable through the adoption of this innovative geometry.

2. Modeling the S-bend flat plate collector

2.1. Geometric model

Heat transfer in a straight tube flat plate collector is typically lower due to its simpler flow pattern compared to that of a bent tube. Here a S-bend tube is employed based on the field synergy theory of a curved tube for enhanced heat transfer (Yang and Tao, 2006) as shown in Fig. 1. The curved tube flat plate collector consists of insulation, absorber plate, a transparent cover and a curved tube connected to the absorber plate. The absorber is coated with a selective with high absorption and low emissivity. For the transparent cover, a high-transmittance glass coated with an antireflection film is used.

2.2. Numerical model for field synergy heat transfer enhancement

Describing the collector in 2D with a rectangular frame, the convective heat transfer equation of the fluid can be written as follows (Jiandong et al., 2015; Guo et al., 1998; Xia, 2001):

$$q^* = \nabla \cdot (K_F \nabla T_F) = \rho c_p \left(u \frac{\partial T_f}{\partial x} + v \frac{\partial T_f}{\partial y} \right) \quad (1)$$

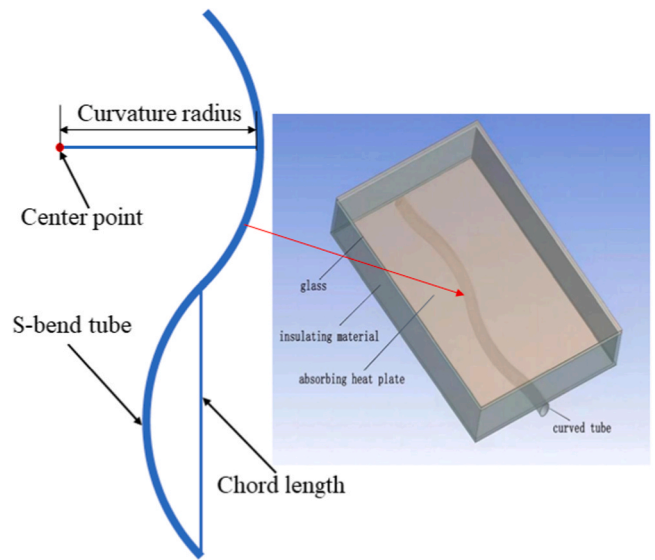


Fig. 1. Geometric model of bends tube.

where K_F represents the thermal conductivity of wall solids, T_F denotes the inner wall temperature, T_f is the water temperature, c_p stands for the specific heat capacity of water, and ρ represents the density of water.

The thermal flux density (q) is determined by integrating Eq.(1) (Jiandong et al., 2015; Guo et al., 1998; Xia, 2001):

$$q = -K_F \frac{dT_F}{dy} \Big|_{y=0} = \int \rho c_p \left(u \frac{\partial T_f}{\partial x} + v \frac{\partial T_f}{\partial y} \right) dy \quad (2)$$

where y represents the direction of the temperature gradient, and $\frac{\partial T_f}{\partial x} = 0$ at the solid-liquid interface ($y=0$), $V = |\vec{U}| \cos \beta$, β is the angle between the fluid velocity ($\vec{U}$) and the temperature gradient ($\frac{\partial T_f}{\partial y}$).

The total absorbing heat (Q) per unit area can be written as (Jiandong et al., 2015; Guo et al., 1998; Xia, 2001):

$$Q = -K_F \frac{dT_F}{dy} \Big|_{y=0} = \rho c_p \int \left(|\vec{U}| \cos \beta \frac{\partial T_f}{\partial y} \right) dy \quad (3)$$

Heat transfer per unit area can be enhanced by increasing the temperature gradient ($\frac{\partial T_f}{\partial y}$), increasing the velocity to reach an optimum flow velocity, influencing ρc_p through material selection, and adjusting β to approach zero ($\cos \beta=1$). In order to improve the synergy angle β , velocity vector, and temperature gradient, this paper applies field synergy theory to optimize the channel shape. Fig. 1 illustrates the adjustment of the flow field direction by altering the chord length and radius to minimize the angle β , achieved through the use of S-bends (Jiandong et al., 2015; Xia, 2001).

3. Heat and mass transfer model and boundary conditions

To investigate the heat transfer mechanism of the flat plate collector, various governing equations were utilized across different structural regions. The continuity, energy, momentum and k- ϵ equations were employed in the fluid zone including the flow field in air and elbow area due to the high Reynolds number and Rayleigh number causing turbulence (Tian and Barigou, 2015; Franco and Bartoli). The thermal conductivity equation was employed in non-fluid areas such as glass, thermal insulation material, heat absorption plate and bends tube. The k- ϵ turbulence governs the convective heat transfer as follows: The specific development of k- ϵ equations in this case is as follows:

Continuity equation (Qiu et al., 2015; Gong et al., 2020):

$$\frac{\partial}{\partial x_i}(\rho u_i) = 0 \quad (4)$$

Energy equation (Qiu et al., 2015; Gong et al., 2020) :

$$\frac{\partial}{\partial x_i}(\rho u_i T_f) = \frac{\partial}{\partial x_i} \left[\left(\frac{\mu}{Pr} + \frac{\mu_t}{\sigma_T} \right) \frac{\partial T_f}{\partial x_i} \right] + s_h \quad (5)$$

Momentum equation (Qiu et al., 2015; Gong et al., 2020) :

$$\frac{\partial}{\partial x_j}(\rho u_i u_j) = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} \left[(\mu_t + \mu) \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \frac{2}{3} \frac{\partial u_k}{\partial x_k} \delta_{ij} \right) - \frac{2}{3} \rho k \delta_{ij} \right] + \rho g_i \quad (6)$$

k-ε equations (Qiu et al., 2015; Gong et al., 2020):

$$\frac{\partial}{\partial x_i}(\rho u_i k) = \frac{\partial}{\partial x_i} \left[\left(\mu + \frac{\mu_t}{\sigma_k} \right) \frac{\partial k}{\partial x_i} \right] + G_k - \rho \epsilon \quad (7)$$

$$\frac{\partial}{\partial x_i}(\rho u_i \epsilon) = \frac{\partial}{\partial x_i} \left[\left(\mu + \frac{\mu_t}{\sigma_\epsilon} \right) \frac{\partial \epsilon}{\partial x_i} \right] + \frac{\epsilon}{k} \left(c_1 G_k - c_2 \rho \epsilon \right) \mu_t = c_u \rho \frac{k^2}{\epsilon} \quad (6')$$

$$G_k = \mu_t \frac{\partial u_i}{\partial x_j} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \quad (8)$$

σ_ϵ and σ_k represent the turbulent Prandtl numbers governing the turbulent diffusion of ϵ and k , which are set to 1.3 and 1.0; the constants c_u , c_1 , and c_2 are assigned values of 0.09, 1.44, and 1.92, respectively; S_h denotes the heat source term; g_i is the acceleration due to gravity; λ_c stands for the thermal conductivity (Gong et al., 2020).

The boundary conditions are the following:

- 1) The working fluid is selected as water, and the relevant parameters are directly called from the Fluent database with corresponding constants. The inlet flow velocity of the collector flow channel was fixed at 0.03 m/s, while the outlet was designed as outflow since the temperature and pressure of the simulated and target values are unknown.
- 2) The surface heat flux density of the heat absorbing plate is obtained by SolTrace optical simulation as shown in Fig. 2, and the emissivity (ϵ_{abs}) of the surface coating is 0.08 (Refer to Table 1).
- 3) The temperature variation falls within a limited range. To streamline calculations and enhance the stability of gas density and airflow field, the Boussinesq model is employed.
- 4) This simulation example utilized a plate collector measuring 0.2 m in length and 0.15 m in width, and all material-related parameters are shown in Table 1.
- 5) S2S radiation model is used to simulate radiant heat exchange in the air region, taking into account only the heat radiation between the

Table 1

Geometric, optical and thermo-physical parameters (Ceron et al., 2015; Gong et al., 2020).

Parameters	Value
Inside radius of tube, mm	9
Outside radius of tube, mm	10
Length of plate collector, m	0.2
Width of plate collector, m	0.15
Absorptivity of the coating, -	0.96
Chord length of S-bend, mm	80
Emissivity of the coating, -	0.08
Density of insulation material, kg/m ³	96
Thermal conductivity of insulation material, W/m-K	0.023
The specific heat capacity of insulation material, J/kg-K	80
Emissivity of glass, -	0.86
Transmissivity of glass, -	0.96
Density of glass, kg/m ³	2230
Thermal conductivity of glass, W/m-K	1.2
The specific heat capacity of the glass, J/kg-K	837
Density of copper, kg/m ³	8978
Thermal conductivity of copper, J/kg-K	381
The specific heat capacity of copper, W/m-K	387.6

surfaces of the model. The surface of the insulating material and the glass is set as a mixed boundary of thermal radiation and convection.

- 6) Radiation and convection are the main forms of heat transfer in the system, in which the insulation material takes heat from the heat absorbing plate and dissipates it into the environment in the same form. The energy loss equation is as follows (Alshukri et al., 2022):

$$Q_{\text{loss}} = S \bullet [h_{a-g}(T_w - T_a) + \sigma \epsilon_g (T_w^4 - T_s^4)] \quad (9)$$

where T_w is the temperature of the wall; T_a is the temperature of air and in this case is equal to 300 K; T_s signifies the sky temperature which was set to be 287 K; h_{a-g} stands for the external convective heat transfer coefficient; ϵ_g is the emissivity of glass; σ is the Stefan-Boltzmann constant ($5.67 \times 10^{-8} \text{ W} \cdot \text{m}^{-2} \cdot \text{K}^{-4}$); S indicates the surface area of heat radiation and convection to surrounding environment. Considering that most of the flat plate collectors are placed on the roof, the velocity of wind is about 3–4 m/s setting then h_{a-g} to 5 W/m²·K in this paper (Alshukri et al., 2022).

The solar radiation hits the flat-plate collector at the incidence angle and is absorbed by the selective coating. A fraction of the heat is subsequently dissipated through thermal radiation and convection, while the remaining portion is absorbed by the fluid in the tube beneath the plate, thereby becoming useful heat. To obtain the surface heat flux needed in the thermal analysis as a boundary condition, the SolTrace software was for the optical analysis needed (Alshukri et al., 2022; Wendelin et al., 2013). The optical parameters listed in Table 1 were utilized in SolTrace for optical simulation, resulting in the surface heat flux density at the absorbing plate as depicted in Fig. 2. It is worth noting that positive and negative values appear from the midpoint onwards.

Under the standard condition of DNI=1000 W/m², the surface of the heat absorber plate is evenly distributed with a heat flux of 921 W/m² (Fig. 2) after passing through the glass with a transmittance of 0.96 and the coating with an absorption of 0.96.

4. Verification of the thermal model

To ensure the validity and precision of the proposed model, the convective terms in Eqs. (4)–(8) were discretized using a first-order upwind scheme. The pressure term was determined by the weighted body force, while the momentum term was handled using a first-order upwind scheme. The velocity-pressure coupling was achieved using the COUPLE algorithm. Initially, the air density was set to 1.1405 kg/m³ based on the Boussinesq model outlined earlier, with a coefficient of thermal expansion set to 0.00344827 K⁻¹. The governing equations were solved using Fluent 18.0 employing the finite volume method (Nima and

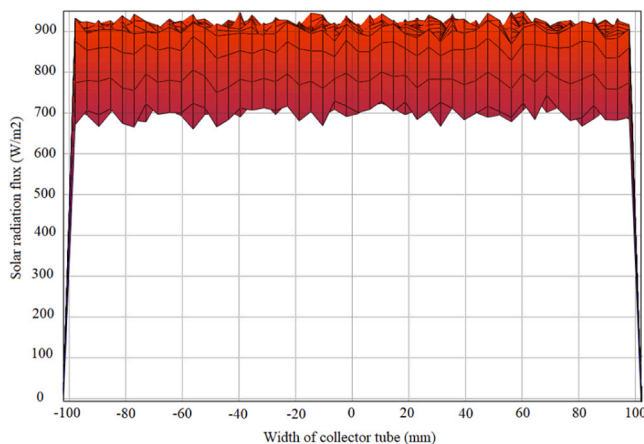


Fig. 2. Surface heat flow density of the absorbing heat plate.

Ali, 2017).

The grid layout for the entire collector is illustrated in Fig. 3, where the plate space is meshed using hexahedral meshing and geometrically complex areas are meshed using tetrahedral meshing. To ensure that the calculated results remain robust against variations in mesh density, thereby verifying mesh independence, sensitivity analyses were conducted. The curved tube flat plate collector, with a radius of curvature of 90 mm, employs the average temperature of the heat absorption plate (T_{abs}) and output energy as criteria for assessing goodness. The results, detailed in Table 2, demonstrate that the relative change in T_{abs} is less than 0.001 K, and the output energy change is 0.001 W when the element size is adjusted from 0.5 mm to 0.25 mm. Consequently, a mesh system with an element size of 0.5 mm was selected for the model, striking a balance between computational efficiency and accuracy.

When $T^*=0$ in the references (Cerón et al., 2015; García et al., 2013), the thermal efficiency obtained in the experiment is about 79 % (Cerón et al., 2015; García et al., 2013). And by inputting its corresponding parameters into this model, the final thermal efficiency obtained through calculation is 78 %. The numerical calculation results are almost consistent with the experimental results and less than 1 %, indicating satisfactory accuracy and the correctness of the model. Explanation: T_m = mean fluid temperature, T_a = ambient temperature (the temperature of air and in this case is equal to 300 K), and $T^*=(T_m-T_a)/DNI$.

5. Results

5.1. Thermal analysis

The collector simulations were done with the surface temperature values given in Fig. 2 as thermal boundary conditions, and with the geometric dimensions and thermal parameters in Table 1. The simulations were done with the FLUENT numerical simulation software (Wendelin et al., 2013; Liu et al., 2016). In the next, the main results of the thermal analysis comparing the two collector geometries are presented.

The simulated temperature distributions on the surface of the absorbing heat plate are depicted in Fig. 4. In the case of straight tubes, the temperature distribution appears symmetrical, whereas with S-bend tubes, the isotherms exhibit an S-shaped pattern reflecting the curvature of the tubes. As one moves farther from the flow field, temperatures tend to increase in both scenarios due to greater thermal resistance over longer distances, hindering heat absorption by the fluid and leading to its accumulation in the plate. Furthermore, the temperature distribution on the absorbing heat plate reveals a gradual convergence of fluid temperature towards the plate temperature, resulting in higher temperatures near the outlet.

The collector model depicted in Fig. 4 is relatively small (Refer to Table 1 for the data), resulting in an absorbed solar energy of only 27.6 W. Consequently, the differences observed between the depicted cases are minute. The curvature radius of the S-bend tubes has only a slight impact on the average temperature of the absorbing heat plate: A curvature radius of 97.5 mm yielded an average temperature of 314 K, while a radius of 82.5 mm resulted in 314.1 K, with an 80 mm chord

Table 2

The test of mesh independence.

Element size (mm)	Average temperature of the heat absorption plate (K)	Energy absorbed by fluid (W)
0.5	313.886	25.707
0.25	313.887	25.706

length, indicating a difference of merely 0.9 K. Interestingly, as the curvature radius of the S-bend increases, the average temperature of the absorbing heat plate initially decreases before subsequently increasing. Notably, the mean temperature of the absorbing heat plate was lowest with a 90 mm curvature radius S-bend tube, measured at 313.9 K.

Therefore, the S-bend flat collector with a curvature radius of 90 mm was selected for comparison with the traditional straight-tube flat collector shown in Fig. 5.

Fig. 5 shows that the surface temperature of the absorbing heat plate decreases by more than 0.3°C owing to the improved heat transfer facilitated by the S-bend configuration, resulting in reduced external radiation losses and increased heat absorption by the fluid. Moreover, the temperature difference between the inlet and outlet increases by 0.02°C (0.6 %), while the thermal efficiency experiences a 0.1 % enhancement.

In Fig. 6, the thermal efficiency and temperature differences within the collector are depicted for varying inlet temperatures. As the fluid inlet temperature rises, the temperature disparity between the absorbing heat plate and the fluid diminishes, resulting in a decrease in the fluid's temperature rise within the tube and consequently reducing thermal efficiency. This decline is attributed to higher surface temperatures, which amplify heat losses as per Eq. (9).

The fluid temperature, which ranged from 300 K (27°C) to 355 K (82°C), experienced a 55°C increase. Consequently, the temperature difference between the inlet and outlet decreased from 3.4°C to 2.5°C, equivalent to a 26.5 % reduction in absorbed energy. Consequently, thermal efficiency also declined from 93.1 % to 69.08 %, with the temperature difference between the absorbing heat plate and the fluid decreasing from 14°C to 10°C.

As depicted in Fig. 6, the fluid within the S-bend concentrator, featuring a curvature radius of 90 mm, operates within the temperature range of 300 K (27°C) to 355 K (82°C). The average thermal efficiency is measured at 80.7 %, marking a 0.1 % increase compared to the collector with straight tubes. This enhancement is attributed to the decrease in the temperature of the heat-absorbing plate by 0.33°C, facilitated by the strengthened heat transfer due to the S-bend configuration (Refer to Fig. 6).

To estimate the heat transfer improvement of S-bend, the modified heat transfer performance factor (η_{HTP}) is defined as follows:

$$\eta_{HTP} = \frac{(Nu/Nu_0)}{(f/f_0)^{\frac{1}{3}}} \quad (10)$$

$\eta_{HTP} > 1$ indicates that the overall performance is improved by adding fins; $\eta_{HTP} < 1$ means that increasing pumping power requirement exceeds the thermal enhancement (Gong et al., 2017).

The friction factor (f) is calculated as (Bellos et al., 2017, 2018):

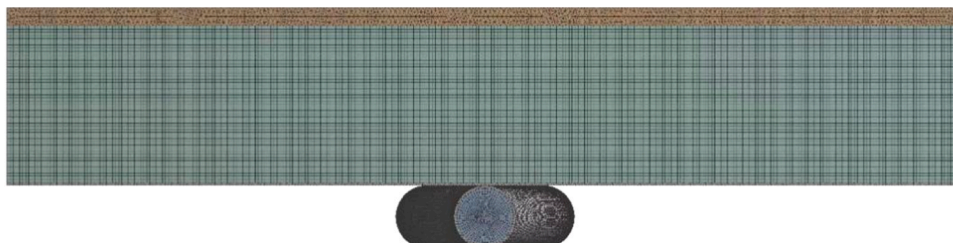


Fig. 3. Schematic diagram of the flat plate collector grid.

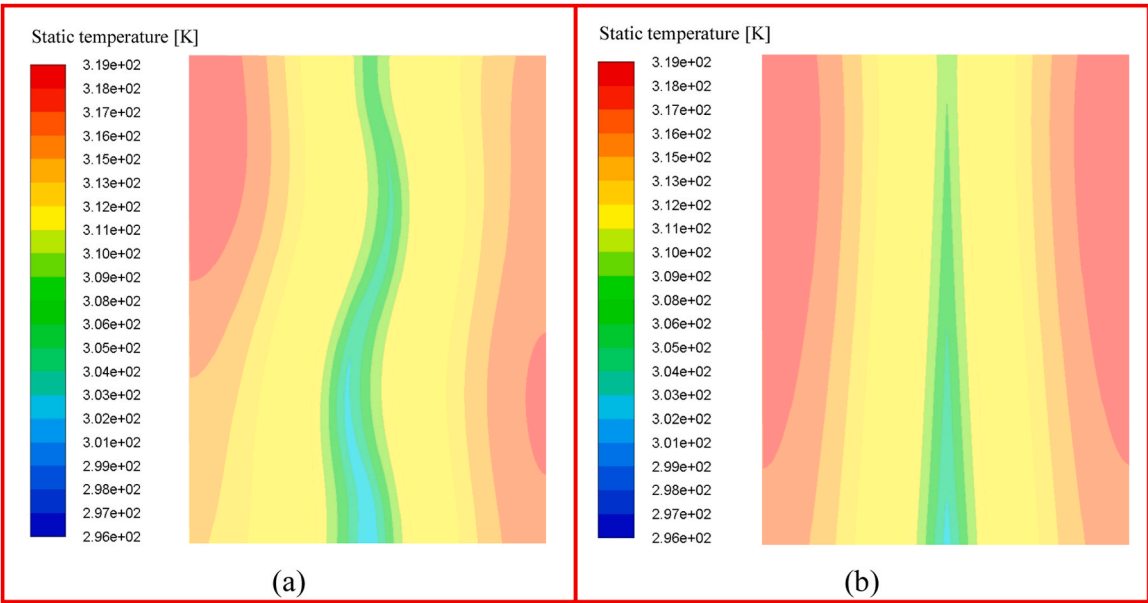


Fig. 4. Temperature distribution on the surface of absorbing heat plate, (a) S-bend, (b) straight-tube.

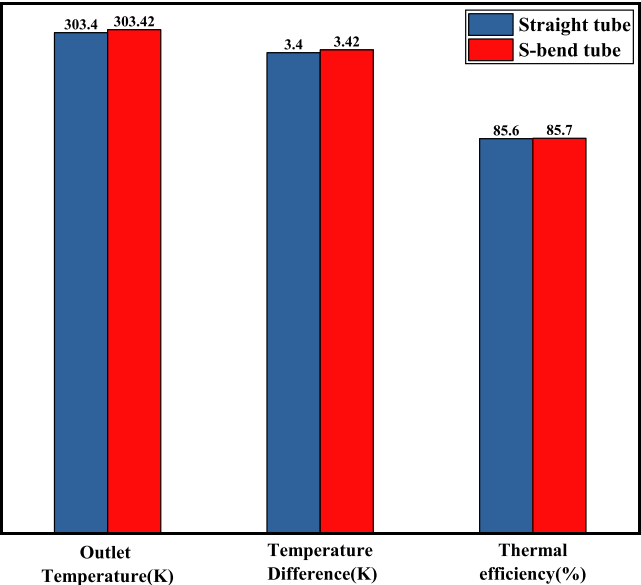


Fig. 5. Comparison between S-bend tube and straight tube.

$$f = \frac{\Delta p}{\frac{1}{2} \rho \times U^2} \times \left(\frac{D}{L}\right) \tag{11}$$

Among them, Δp represents the pressure drop through the channel, ρ is the density of the working fluid (water), D and L are the diameter and length of the channel, respectively. By calculating these results, Table 3 was obtained for further data interpretation. The calculation result is greater than 1, indicating that S-bend has improved overall performance.

5.2. Flow field analysis in the tube

The S-bend tube enhances heat transfer, resulting in increased thermal efficiency and reduced temperature of the heat-absorbing plate. To delve deeper into the factors driving this enhanced heat transfer, the flow field within the tube is analyzed next. Fig. 7 displays the magnitude of the flow field, while Fig. 8 illustrates the product of the velocity vector

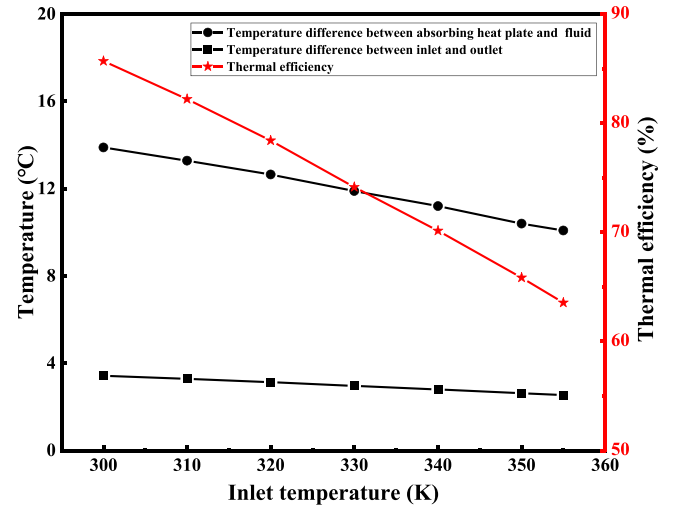


Fig. 6. Comparing a straight-tube and S-bend (90 mm) collector at different temperatures.

Table 3
Calculation of the heat transfer performance factor(η_{HTP}).

Type	Δp (Pa)	f	Nu	η_{HTP}
S-bend	4.97	0.452	89.39	1.05
straight-tube	4.67	0.467	85.77	

and the temperature gradient.

As shown in Fig. 7, the flow velocity in the straight tube exhibits a symmetric distribution, with higher flow rates observed in the central region and lower flow rates towards the periphery. Conversely, in the S-bend configuration, the region of maximum flow velocity undergoes variation, shifting closer to the outer region of the bend. With an inlet flow velocity set at 0.03 m/s, the maximum flow velocity measures 0.0514 m/s in the S-bend tube and 0.0512 m/s in the straight tube. Consequently, alterations in flow velocity and the location of peak velocities impact heat transfer dynamics. In line with the field synergy heat transfer enhancement described in Eq. (3), the S-bend configuration

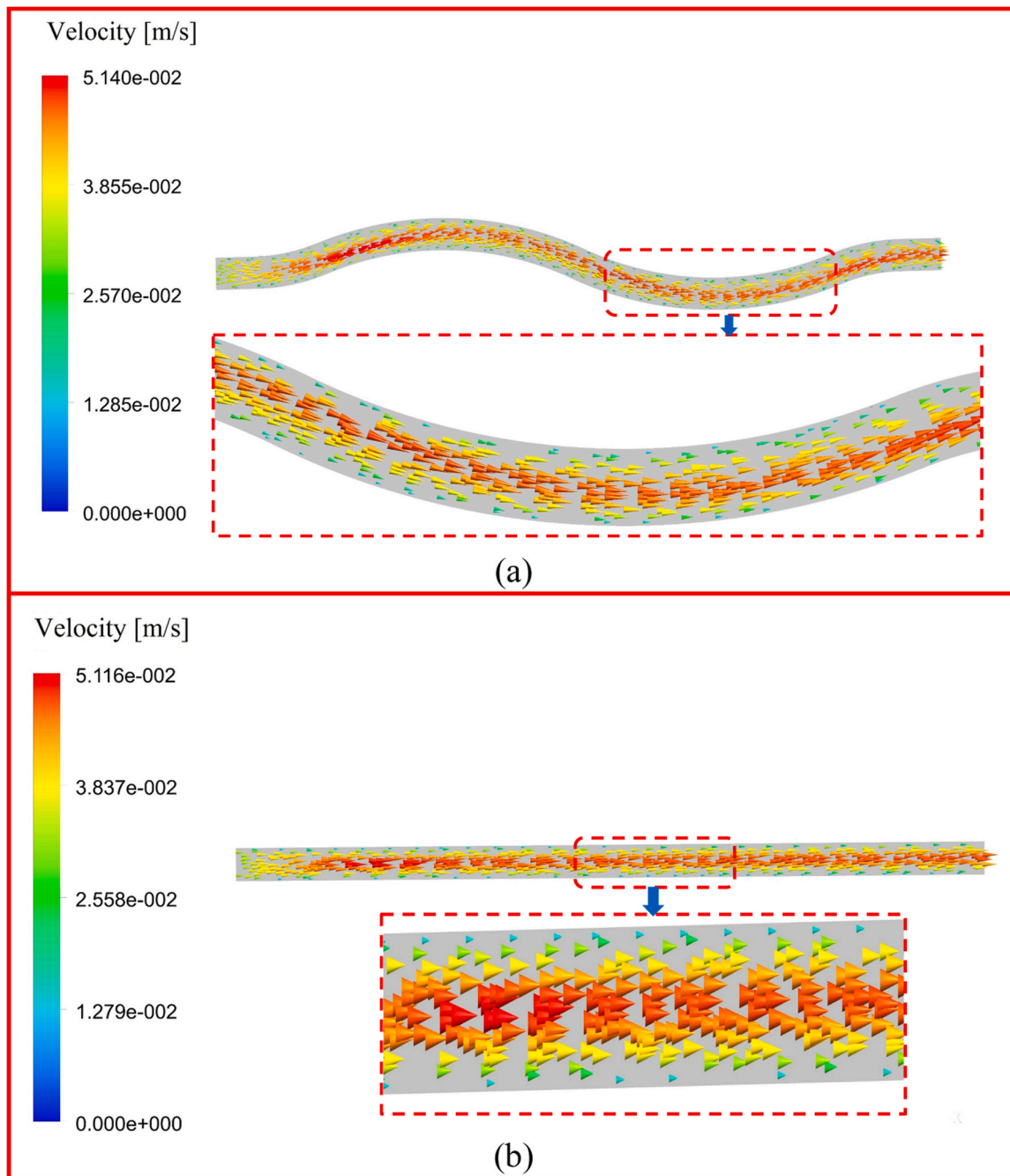


Fig. 7. Velocity in flow field in the tube. (a) S-bend, (b) straight-tube.

alters both the temperature gradient and the velocity vector, thereby reducing β between the temperature gradient and the velocity vector, effectively increasing the cosine of β . This effect is visually depicted in Fig. 8.

In Fig. 8, at the entrance of the two tubes, there is an entrance effect of fluid transition from laminar flow to turbulent flow which makes the velocity and direction of the fluid change and β between the velocity and temperature gradient also change, resulting in a large value in their product as shown in Fig. 8. At the same time, it also shows that heat transfer is enhanced at the entrance absorbing more heat. The main reason for this is that the wall resistance reduces the flow velocity close to the inner wall of the tube and increases the pressure. In accordance with the flow conservation law, the flow velocity increases in the intermediate region while the intermolecular forces diminish. The force exerted by the inner wall facing the center alters the fluid flow direction

and generates a velocity component along the vertical inner wall. So that, the angle β between the fluid velocity and the temperature gradient approaches zero, leading to an increase in their product. As shown in Fig. 8, this locally enhances heat transfer, as described by Eq. (3).

For the straight tube, the fluid remains in a turbulent yet stable state, reflected in Fig. 8(b) where the product of velocity and temperature gradient remains consistent. This is characterized by a high value in the central region and a narrower, lower value towards the edges. This distribution is primarily due to the higher fluid velocity in the central area compared to the slower velocity towards the edges. Conversely, in the case of the S-bend tube, the flow state undergoes continuous change due to the bending, leading to fluctuations in the product of temperature and velocity gradient, as illustrated in Fig. 8(a). Local strengthening occurs particularly at the bend, with a significant increase in the product between fluid velocity and temperature gradient. This phenomenon can

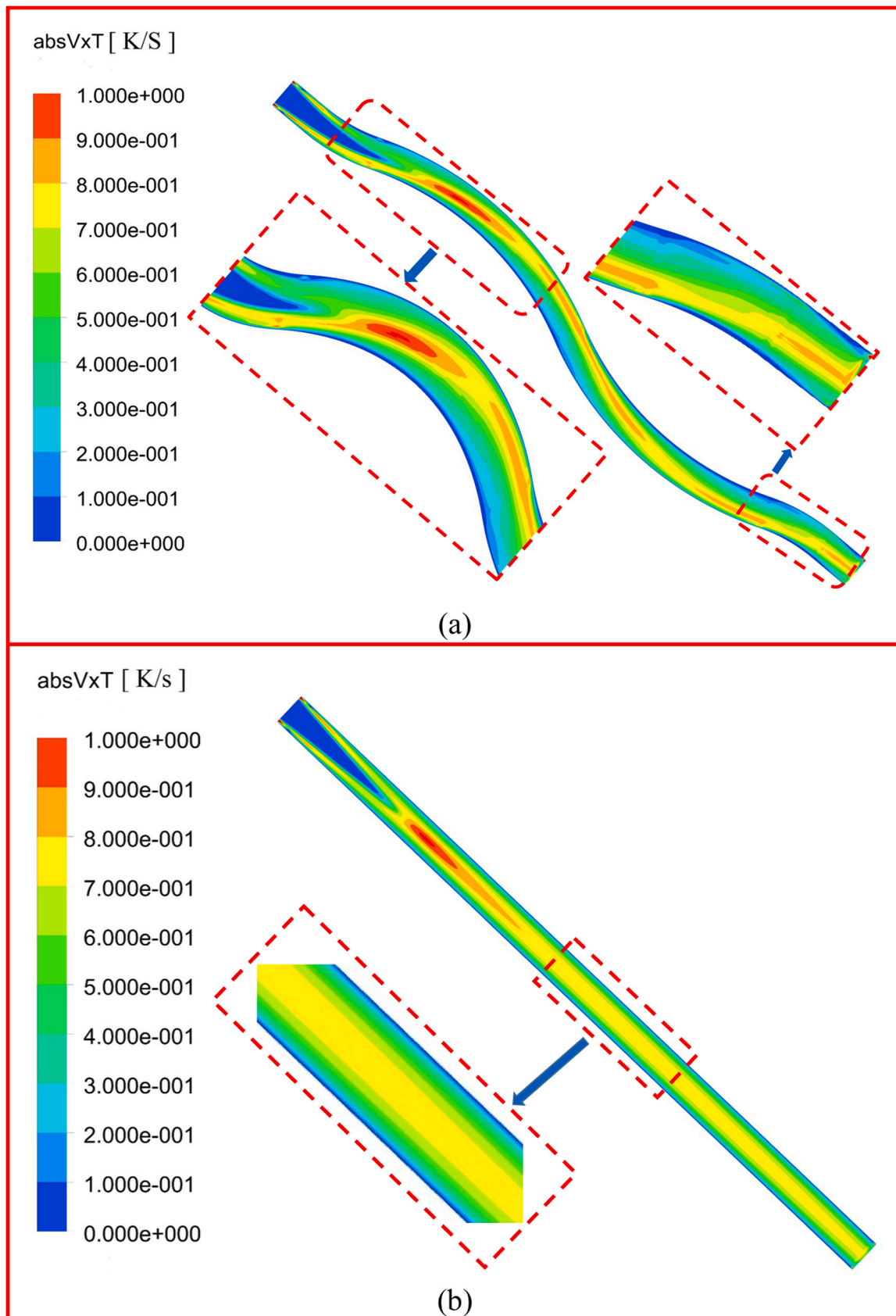


Fig. 8. Product of temperature gradient and velocity vector. (a) S-bend, (b) straight-tube.

be attributed to the presence of a centrifugal force at the bend, which causes the fluid to flow towards the outer side. As a result, the pressure decreases at the inner side, directing the velocity vector towards the inner side.

5.3. Analysis of air flow characteristics in cavity

The temperature difference between the absorbing heat plate and the glass cover causes convection in the air space as shown in Fig. 9. The Knudsen number is <0.01 meaning that there is a counter rotating vortex in the air flow field, and the air flow is in a stable thermal convection state.

In the S-bend tube collector model, the airflow exhibits greater complexity attributed to the asymmetric temperature distribution. This is evident in Fig. 9(a), where the flow velocity appears in the vertical section direction with an asymmetric distribution. Conversely, in the straight-tube collector model, characterized by symmetrical internal temperature distribution, the airflow remains relatively regular. In Fig. 9(b), the overall flow velocity is downward and symmetrically distributed. The airflow patterns shown in Fig. 9 play a crucial role in determining the temperature distribution on the glass, as illustrated in Fig. 10.

The temperature distribution on the heat-absorbing plate dictates the airflow pattern within the cavity. As shown in Fig. 4, the lowest temperature is observed at the center of the absorbing heat plate, attributed to the downward flow of air at the center. This temperature gradually increases towards both sides, prompting air movement patterns illustrated in Fig. 9. Additionally, air moves towards the sides of the collector absorber plate due to lower temperatures resulting from radiative losses to the surroundings, resulting in the formation of four vortices, as shown in Fig. 9.

The airflow state depicted in Fig. 9 influences the energy absorbed by the glass and the resulting temperature distribution, as demonstrated in Fig. 10. The presence of an asymmetric and irregular vortex structure in the S-bend configuration produces an irregular temperature distribution (Fig. 10(a)), whereas the straight-tube configuration yields a symmetrical temperature distribution (Fig. 10(b)). In areas where warm air flows, temperatures are higher, while areas distant from the warm gas exhibit lower temperatures. This phenomenon can be attributed to the β angle, known as field synergy. Finally, a butterfly-shaped temperature distribution diagram appears in Fig. 10, corresponding to the motion mode of the warm airflow cloud depicted in Fig. 9.

Optimizing the convection phenomenon generated inside the air chamber is a majorization (the emergence of vacuum heat pipes is

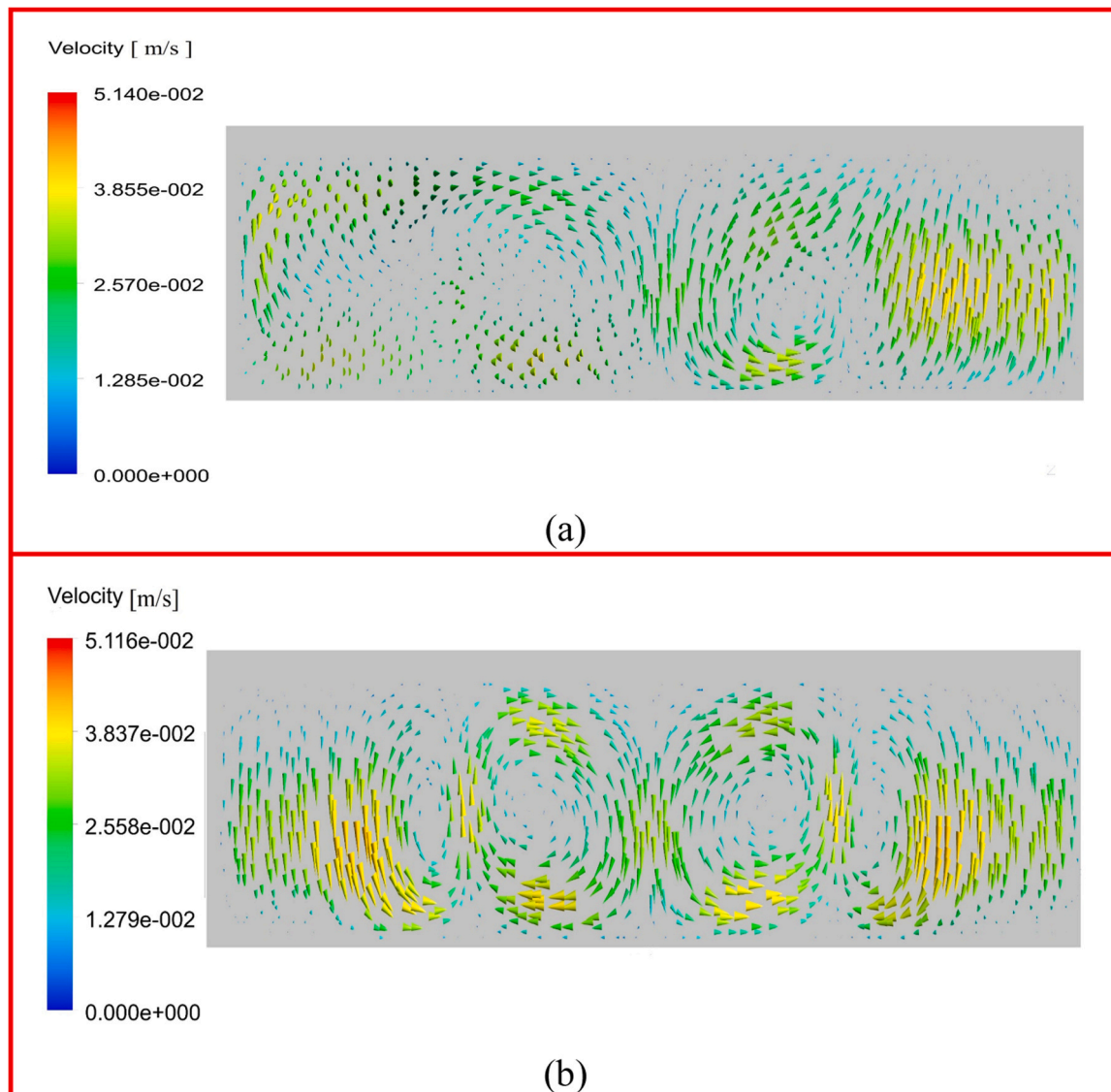


Fig. 9. R-B heat convection of air in the chamber. (a) S-bend, (b) straight-tube.

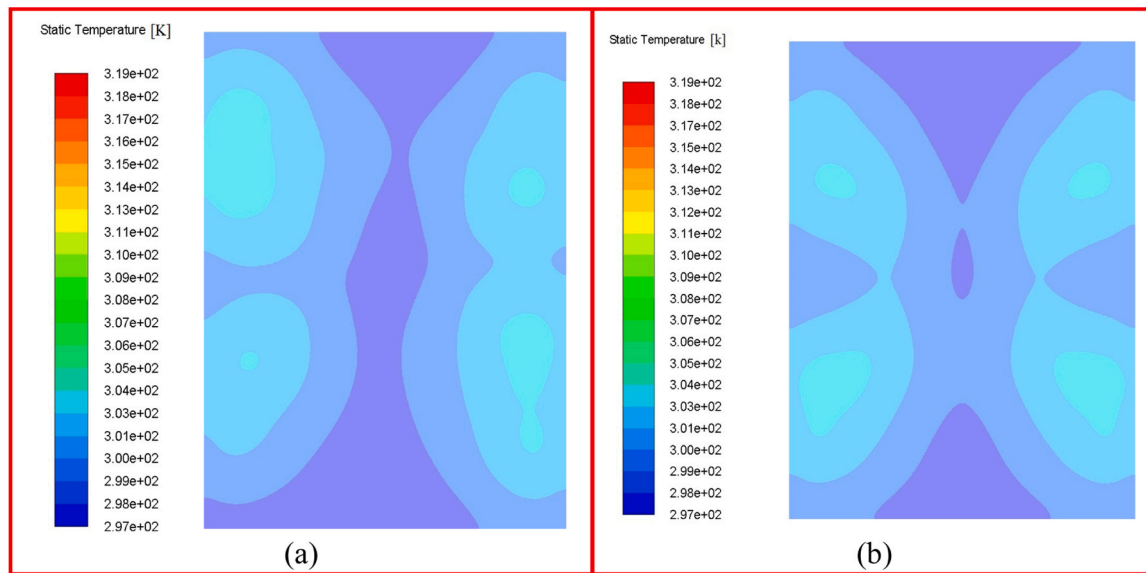


Fig. 10. Glass surface temperature. (a) S-bend, (b) straight-tube.

closely related to it). Observing Fig. 9, it is not difficult to find that the use of S-bend not only enhances heat transfer, but also to some extent destroys the vortex structure generated by air convection. When the average flow velocity inside the cavity does not change significantly and is maintained at around 0.015 m/s, the vortex structure generated by the destruction can to some extent suppress the convective phenomenon formed. The vector diagram shown in Fig. 9 to some extent indicates that this study indirectly weakens the convection phenomenon by changing the flow channel form. The fundamental reason is that the temperature distribution on the upper and lower plate surfaces changes, leading to disturbances in the driving factors. This provides certain ideas and references for subsequent work. By indirectly suppressing convection by changing the distribution of temperature or directly suppressing convection through other methods such as reasonable arrangement of TIM, equipment optimization and efficiency improvement can be achieved.

This article presents the results and analysis of numerical modeling, but due to limitations in site and external factors, it is not possible to conduct further comprehensive experimental evaluation of the S-bend collector. We have to admit this limitation, and the potential challenges faced by experimental research on composite blocks are the production of S-bends and how to accurately feedback the vortex structure inside the cavity for further tracking and research under relatively low flow rates. This study not only enhances heat transfer, but also aims to weaken convection in the cavity area. By changing the component model, this goal is achieved, providing a train of thought for subsequent research.

6. Conclusion

In this paper, an S-bend collector tube form is used as the flow channel system in flat plate solar collector to enhance heat transfer in the collector. A set-up model with 0.2 m long and 0.15 m wide and 80 mm chord length S-bend was used to numerically analyze the effects on the efficiency of the collector, heat transfer and flow state in the tube, and distribution and motion mode of air in the cavity between the absorber and glass cover. The simulated numerical growth of a single block (0.2 m long and 0.15 m wide) is not very significant, but in the case of large flat plate combinations, corresponding benefits can be obtained and can be coupled with other enhanced heat transfer methods. (Due to limitations in computing power and external conditions, simulating the entire tablet is not cost-effective, so we chose to simulate a single block.)

The main conclusions are as follows:

- An S-bend collector with a 90 mm curvature radius demonstrated a 0.1 % improvement in thermal efficiency compared to a traditional straight-tube collector, accompanied by a reduction in the absorber plate temperature by 0.33°C. Meanwhile, the heat transfer performance factor (η_{HTP}) is greater than one.
- The optimization reduces the angle (β) between the fluid velocity vector and temperature gradient, approaching zero, thus enhancing heat transfer in accordance with the field synergy theory.
- Within the air cavity between the collector absorber and glass cover, the straight-tube collector produces a symmetric and relatively uniform vortex structure, while the S-bend configuration leads to a complex and irregular vortex structure at almost the same flow rate.
- A butterfly-shaped temperature distribution emerges on the glass, corresponding to the motion pattern of the warm airflow velocity. The average glass temperature of the S-bend type collector is slightly lower than that of the straight-tube collector.

The air in the cavity has a vortex structure and the temperature distribution on the glass shows a butterfly-shape. These new phenomena are worthy of further study, so as to propose better optimization methods.

CRediT authorship contribution statement

The manuscript has not been published previously and it is not under consideration for publication elsewhere. The publication is approved by all authors and explicitly by the responsible authorities. If accepted, the article will not be published elsewhere in the same form, in English or in other language, without the written consent of the publisher.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

No data was used for the research described in the article.

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